

EXHIBIT 12

Patent Assignment Abstract of Title

Search results for Patent #: 9865546

Generated on Fri Mar 20 2026 16:05:16 GMT-0400 (Eastern Daylight Time)

Invention title	Patent number	Application number	Publication number	PCT number
CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME	9865546	14729298	20160358860	N/A
	Issue date	View in PPUBS	View in PPUBS	International number
	Jan 9, 2018			
Inventors		Filing date	Publication date	N/A
Emre Alptekin , Nicolas Breil L., Christian Lavoie, Ahmet Ozcan S., Kathryn Schonenberg T.		Jun 3, 2015	Dec 8, 2016	

Assignment 1

Reel/ Frame

[35779 /0136](#)

Pages

10

Conveyance

ASSIGNMENT OF
ASSIGNOR'S INTEREST

Documents

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Recorded

Jun 3, 2015

Assignors & execution date

[ALPTEKIN, EMRE](#)

May 19, 2015

[BREIL, NICOLAS L.](#)

May 20, 2015

[LAVOIE, CHRISTIAN](#)

May 13, 2015

[View more](#)

Assignee

[INTERNATIONAL BUSINESS
MACHINES CORPORATION](#)

NEW ORCHARD ROAD

ARMONK

NEW YORK, UNITED STATES

10504

Correspondent

HOFFMAN WARNICK LLC

540 BROADWAY

4TH FLOOR

ALBANY, NY 12207

Attorney docket number

FIS920150066US1

Assignment 2

Reel/ Frame

[37409 /0869](#)

Pages

15

Conveyance

ASSIGNMENT OF
ASSIGNOR'S INTEREST

Documents

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Recorded

Jan 3, 2016

Assignors & execution date

[INTERNATIONAL BUSINESS
MACHINES CORPORATION](#)

Oct 28, 2015

Assignee

[GLOBALFOUNDRIES U.S. 2 LLC](#)

2070 ROUTE 52

HOPEWELL JUNCTION

NEW YORK, UNITED STATES

12533

Correspondent

HESLIN ROTHENBERG
FARLEY & MESITI P.C.

5 COLUMBIA CIRCLE

ALBANY, NY 12203

Attorney docket number

3718.266

Assignment 3

Reel/ Frame

[37542 /0087](#)

Pages

18

Conveyance

ASSIGNMENT OF
ASSIGNOR'S INTEREST

Documents

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Recorded

Jan 15, 2016

Assignors & execution date

[GLOBALFOUNDRIES U.S. 2 LLC](#)

Dec 14, 2015

[GLOBALFOUNDRIES U.S. INC.](#)

Dec 8, 2015

Assignee

[GLOBALFOUNDRIES INC.](#)

UGLAND HOUSE

PO BOX 309

GRAND CAYMAN

CAYMAN ISLANDS

KY1-1104

Correspondent

HESLIN ROTHENBERG
FARLEY & MESITI P.C.

5 COLUMBIA CIRCLE

ALBANY, NY 12203

Attorney docket number

3718.266

Assignment 4

Reel/ Frame

[49490 /0001](#)

Assignors & execution date

[GLOBALFOUNDRIES INC.](#)

Correspondent

CT CORPORATION

Pages
892

Nov 27, 2018

4400 EASTON COMMONS WAY
SUITE 125
COLUMBUS, OH 43219

Conveyance
SECURITY AGREEMENT

Assignee

[WILMINGTON TRUST,
NATIONAL ASSOCIATION](#)
1100 NORTH MARKET STREET
WILMINGTON
DELAWARE, UNITED STATES

Attorney docket number
N/A

Documents

[View documents](#)

Recorded
Nov 29, 2018

19801

Assignment 5

Reel/ Frame

54633 /0001

Assignors & execution date

[GLOBALFOUNDRIES INC.](#)

Correspondent

GLOBALFOUNDRIES
SINGAPORE PTE. LTD.

Pages

1245

Oct 22, 2020

Assignee

[GLOBALFOUNDRIES U.S. INC.](#)
2600 GREAT AMERICA WAY
SANTA CLARA
CALIFORNIA, UNITED STATES
95054

60 WOODLANDS INDUSTRIAL
PARK D STREET 2,
SINGAPORE,
738406 SINGAPORE

Conveyance

ASSIGNMENT OF
ASSIGNOR'S INTEREST

Attorney docket number
N/A

Documents

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Recorded

Nov 2, 2020

Assignment 6

Reel/ Frame

54636 /0001

Assignors & execution date

[WILMINGTON TRUST,
NATIONAL ASSOCIATION](#)

Correspondent

BENJAMIN PETERSEN

Pages

911

Nov 17, 2020

Assignee

[GLOBALFOUNDRIES INC.](#)
PO BOX 309, UGLAND HOUSE
MAPLES CORPORATE
SERVICES LIMITED

1460 EL CAMINO
REAL, 2ND FLOOR
SHEARMAN & STERLING LLP
MENLO PARK, CA 94025

Conveyance

RELEASE OF
SECURITY INTEREST

Attorney docket number
37188/13

Documents

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Recorded
Nov 20, 2020

GRAND CAYMAN
CAYMAN ISLANDS
KY1-1104

Assignment 7

Reel/ Frame

56987 /0001

Pages

1045

Conveyance

RELEASE OF
SECURITYINTEREST

Documents

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Recorded

May 12, 2021

Assignors & execution date

[WILMINGTON TRUST,
NATIONAL ASSOCIATION](#)

Nov 17, 2020

Assignee

[GLOBALFOUNDRIES U.S. INC.](#)

440 STONE BREAK
ROAD EXTENSION

MALTA

NEW YORK, UNITED STATES

12020

Correspondent

BENJAMIN PETERSEN

1460 EL CAMINO
REAL, 2ND FLOOR

SHEARMAN & STERLING LLP
MENLO PARK, CA 94025

Attorney docket number

37188/13

Patent Assignment Abstract of Title

Search results for Patent #: 10707167

Generated on Mon Mar 23 2026 13:15:23 GMT-0400 (Eastern Daylight Time)

Invention title	Patent number	Application number	Publication number	PCT number
CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME	10707167	15826939	20180090447	N/A
	Issue date	View in PPUBS	View in PPUBS	International number
	Jul 7, 2020			
Inventors		Filing date	Publication date	N/A
Emre Alptekin , Nicolas Breil L., Christian Lavoie, Ahmet Ozcan S., Kathryn Schonenberg T.		Nov 30, 2017	Mar 29, 2018	

Assignment 1

Reel/ Frame	Assignors & execution date	Correspondent
49490 /0001	GLOBALFOUNDRIES INC.	CT CORPORATION
Pages	Nov 27, 2018	4400 EASTON COMMONS WAY
892	Assignee	SUITE 125
Conveyance	WILMINGTON TRUST, NATIONAL ASSOCIATION	COLUMBUS, OH 43219
SECURITY AGREEMENT	1100 NORTH MARKET STREET	Attorney docket number
Documents	WILMINGTON	N/A
View documents	DELAWARE, UNITED STATES	
Recorded	19801	
Nov 29, 2018		

Assignment 2

Reel/ Frame	Assignors & execution date	Correspondent
54633 /0001	GLOBALFOUNDRIES INC.	GLOBALFOUNDRIES SINGAPORE PTE. LTD.

Pages
1245

Oct 22, 2020

Conveyance
ASSIGNMENT OF
ASSIGNOR'S INTEREST

Assignee

[GLOBALFOUNDRIES U.S. INC.](#)
2600 GREAT AMERICA WAY
SANTA CLARA
CALIFORNIA, UNITED STATES
95054

60 WOODLANDS INDUSTRIAL
PARK D STREET 2,
SINGAPORE,
738406 SINGAPORE

Documents

[View documents](#)

Attorney docket number
N/A

Recorded
Nov 2, 2020

Assignment 3

Reel/ Frame

54636 /0001

Assignors & execution date

[WILMINGTON TRUST,
NATIONAL ASSOCIATION](#)

Nov 17, 2020

Correspondent

BENJAMIN PETERSEN

Pages

911

1460 EL CAMINO
REAL, 2ND FLOOR

Conveyance

RELEASE OF
SECURITY INTEREST

Assignee

[GLOBALFOUNDRIES INC.](#)
PO BOX 309 , UGLAND HOUSE
MAPLES CORPORATE
SERVICES LIMITED
GRAND CAYMAN
CAYMAN ISLANDS
KY1-1104

SHEARMAN & STERLING LLP
MENLO PARK, CA 94025

Attorney docket number
37188/13

Documents

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Recorded
Nov 20, 2020

Assignment 4

Reel/ Frame

56987 /0001

Assignors & execution date

[WILMINGTON TRUST,
NATIONAL ASSOCIATION](#)

Nov 17, 2020

Correspondent

BENJAMIN PETERSEN

Pages

1045

1460 EL CAMINO
REAL, 2ND FLOOR

Conveyance

RELEASE OF
SECURITY INTEREST

Assignee

[GLOBALFOUNDRIES U.S. INC.](#)
440 STONE BREAK
ROAD EXTENSION

SHEARMAN & STERLING LLP
MENLO PARK, CA 94025

Attorney docket number
37188/13

Documents

[View documents](#)

Recorded

May 12, 2021

MALTA

NEW YORK, UNITED STATES

12020

PATENT ASSIGNMENT COVER SHEET

Electronic Version v1.1
Stylesheet Version v1.2

EPAS ID: PAT3380997

SUBMISSION TYPE:	NEW ASSIGNMENT
NATURE OF CONVEYANCE:	ASSIGNMENT
CONVEYING PARTY DATA	
Name	Execution Date
EMRE ALPTEKIN	05/19/2015
NICOLAS L. BREIL	05/20/2015
CHRISTIAN LAVOIE	05/13/2015
AHMET S. OZCAN	05/13/2015
KATHRYN T. SCHONENBERG	05/13/2015
RECEIVING PARTY DATA	
Name:	International Business Machines Corporation
Street Address:	New Orchard Road
City:	Armonk
State/Country:	NEW YORK
Postal Code:	10504
PROPERTY NUMBERS Total: 1	
Property Type	Number
Application Number:	14729298
CORRESPONDENCE DATA	
Fax Number:	(518)449-0047
<i>Correspondence will be sent to the e-mail address first; if that is unsuccessful, it will be sent using a fax number, if provided; if that is unsuccessful, it will be sent via US Mail.</i>	
Phone:	518-449-0044
Email:	ptocommunications@hoffmanwarnick.com, vfleming@hoffmanwarnick.com
Correspondent Name:	HOFFMAN WARNICK LLC
Address Line 1:	540 BROADWAY
Address Line 2:	4TH FLOOR
Address Line 4:	ALBANY, NEW YORK 12207
ATTORNEY DOCKET NUMBER:	FIS920150066US1
NAME OF SUBMITTER:	AMANDA J. SMITH
SIGNATURE:	/Amanda J. Smith/
DATE SIGNED:	06/03/2015
	This document serves as an Oath/Declaration (37 CFR 1.63).

PATENT

Total Attachments: 8

source=FIS920150066US1_Filed_Dec-Assign#page1.tif

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**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

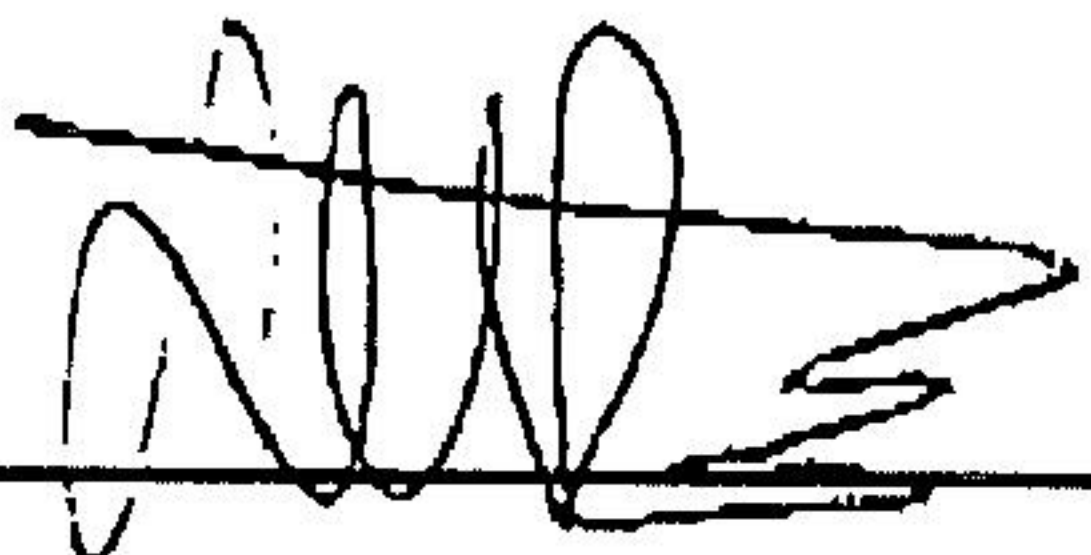
Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature:  Date: 05/19/2015

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

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Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

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I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: Nicolas L. Breil Date: 5/20/2015

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

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(1) Legal Name of Inventor: Emre Alptekin

Signature: _____ Date: _____

(2) Legal Name of Inventor: Nicolas L. Breil

Signature: _____ Date: _____

(3) Legal Name of Inventor: Christian Lavoie

Signature: Christian Lavoie Date: May 13 2015

(4) Legal Name of Inventor: Ahmet S. Ozcan

Signature: _____ Date: _____

(5) Legal Name of Inventor: Kathryn T. Schonenberg

Signature: Kathryn T. Schonenberg Date: May 13, 2015

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
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(1) Legal Name of Inventor: Emre Alptekin

Signature: _____ Date: _____

(2) Legal Name of Inventor: Nicolas L. Breil

Signature: _____ Date: _____

(3) Legal Name of Inventor: Christian Lavoie

Signature: _____ Date: _____

(4) Legal Name of Inventor: Ahmet S. Ozcan

Signature:  _____ Date: 5/13/2015

(5) Legal Name of Inventor: Kathryn T. Schonenberg

Signature: _____ Date: _____

PATENT ASSIGNMENT COVER SHEET

Electronic Version v1.1
 Stylesheet Version v1.2

EPAS ID: PAT3678908

SUBMISSION TYPE:	NEW ASSIGNMENT	
NATURE OF CONVEYANCE:	ASSIGNMENT	
CONVEYING PARTY DATA		
Name		Execution Date
INTERNATIONAL BUSINESS MACHINES CORPORATION		10/28/2015
RECEIVING PARTY DATA		
Name:	GLOBALFOUNDRIES U.S. 2 LLC	
Street Address:	2070 ROUTE 52	
City:	HOPEWELL JUNCTION	
State/Country:	NEW YORK	
Postal Code:	12533	
PROPERTY NUMBERS Total: 182		
Property Type	Number	
Patent Number:	9082851	
Patent Number:	9041079	
Patent Number:	5166552	
Patent Number:	9059139	
Application Number:	14658898	
Application Number:	14710204	
Application Number:	14711119	
Application Number:	14711462	
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Application Number:	14753771
Application Number:	14753682
PCT Number:	CN2014088537
PCT Number:	CN2014089293

CORRESPONDENCE DATA

Fax Number: (518)452-5579

Correspondence will be sent to the e-mail address first; if that is unsuccessful, it will be sent using a fax number, if provided; if that is unsuccessful, it will be sent via US Mail.

Phone: 5184525600

Email: jag@hrfmlaw.com

Correspondent Name: HESLIN ROTHENBERG FARLEY & MESITI P.C.

Address Line 1: 5 COLUMBIA CIRCLE

Address Line 4: ALBANY, NEW YORK 12203

ATTORNEY DOCKET NUMBER:	3718.266
NAME OF SUBMITTER:	JACQUELYN A. GRAFF
SIGNATURE:	/Jacquelyn A. Graff/
DATE SIGNED:	01/03/2016

Total Attachments: 9

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Agreement Reference Number L159424 2008 *smf*

PATENT ASSIGNMENT AGREEMENT ("Agreement") between INTERNATIONAL BUSINESS MACHINES CORPORATION, a New York corporation ("Assignor"), and GLOBALFOUNDRIES U.S. 2 LLC., a Delaware limited liability company ("Assignee"), having an effective date that is the date this Agreement has been executed by the last party to duly sign (the "Effective Date").

WHEREAS, Assignor and GLOBALFOUNDRIES U.S. Inc. have entered into the MASTER TRANSACTION AGREEMENT, dated October 18, 2014, between Assignor and GLOBALFOUNDRIES U.S. Inc. ("Master Transaction Agreement");

WHEREAS, pursuant to the Master Transaction Agreement, Assignor has agreed to sell, transfer, convey, assign and deliver to GLOBALFOUNDRIES U.S. Inc. or its designated affiliate(s), and GLOBALFOUNDRIES U.S. Inc. or its designated affiliate(s) have agreed to accept and acquire all of Assignor's rights, title and interests in and to the patent applications and patents listed in Exhibits A, B, and C and other associated rights as described below;

WHEREAS, GLOBALFOUNDRIES U.S. Inc.'s designee for the purpose of Assignor's assignment of such patent applications and patents hereunder is Assignee; and

WHEREAS Assignee has executed a joinder agreement to the Master Transaction Agreement.

NOW, THEREFORE, for good and valuable consideration, the receipt of which is hereby acknowledged:

1. Subject to and in accordance with the Master Transaction Agreement (including, without limitation, Section 2.1 and the rights, reservations and obligations set forth in Sections 2.10(e)-(g)), Assignor hereby grants, transfers, conveys, delivers and assigns to Assignee any and all worldwide rights, title and interest Assignor holds, or may come to hold, in and to (a) the United States patents identified in Exhibit A attached hereto, the foreign patents identified in Exhibit B attached hereto and the United States and foreign patent applications identified in Exhibit C attached hereto, (b) any other patent or patent application that is a docket family member of any of the patents or patent applications identified in Exhibit A, Exhibit B or Exhibit C, but only to the extent that such other patent or patent application was not previously assigned to Assignee according to the Master Transaction Agreement, (c) all rights relating to any of the foregoing in (a) and (b) that are provided for in the Master Transaction Agreement, (d) all claims, causes of action, rights of recovery, credits and rights of set off of any kind, including, without limitation, all claims for infringement, misappropriation or other violation, with respect to any of the foregoing in (a), (b) and (c), and (e) all payments, income, and royalties and rights to payments, income, and royalties, in each case arising out of the sale, lease, license, assignment, or other disposition thereof after the Effective Date (collectively, "Assigned Patents"). For the purposes of applying the relevant sections of the Master Transaction Agreement to this Agreement as specified in the first sentence of this Section 1, the term "Closing Date" as used in the Master Transaction Agreement is hereby redefined to instead have the meaning assigned in this Agreement to the Effective Date.
2. Assignor and Assignee hereby request and authorize the relevant authority at the United States Patent and Trademark Office and/or the applicable foreign authorities to record Assignee as the assignee and owner of the Assigned Patents.

3. Assignor hereby covenants and agrees that, at any time and from time to time after the date of this Agreement, at Assignee's request, Assignor shall reasonably cooperate with Assignee to execute and deliver such documents and other papers, and otherwise provide reasonable assistance to Assignee, as may be required to grant, transfer, convey, deliver and assign to Assignee any of the Assigned Patents; provided, however, that any formal assignment document recorded with a patent office or other governmental authority shall reflect that such assignment is subject to the rights, reservations and obligations set forth in the Master Transaction Agreement. Assignee shall bear all reasonable out-of-pocket expenses incurred by Assignor in connection with providing the foregoing cooperation and assistance described in this Section 3.
4. It is understood that any finding of invalidity of one assignment as effected hereby shall not affect the assignment of other Assigned Patents.
5. Other than the fact that the term Closing Date as used in the Master Transaction Agreement has been redefined by this Agreement to instead have the meaning assigned in this Agreement to the Effective Date for purposes of interpreting the provisions of Section 1 above including the rights, reservations and obligations set forth in Sections 2.1 and Sections 2.10(e)-(g) of the Master Transaction Agreement, nothing contained in this Agreement shall in any way supersede, modify, replace, amend, change, rescind, waive, exceed, expand, enlarge or in any way affect any of the provisions (including, without limitation, the warranties, covenants, agreements, conditions, representations or, in general, any of the rights and remedies) or any of the obligations set forth in the Master Transaction Agreement, nor shall this Agreement expand or enlarge any remedies under the Master Transaction Agreement.
6. This Agreement is binding upon and inures to the benefit of Assignor and Assignee and their respective successors and assigns. This Agreement and the rights and obligations of Assignor and Assignee under this Agreement shall be governed by, and construed in accordance with the laws of the State of New York.
7. This Agreement and any amendments hereto may be signed in one or more counterparts, each of which, when signed and delivered, shall be deemed to be an original. All such counterparts together shall constitute one and the same valid and binding agreement, even if all of the parties have not signed the same counterpart. Signatures to this Agreement may be delivered by facsimile or other electronic transmission, in which case the facsimile or other electronic copy of an original signature shall be deemed to be an original signature.

[Signature Page Follows]

IN WITNESS WHEREOF, Assignor and Assignee have caused this Agreement to be duly executed in duplicate originals by their duly authorized representative as of the day and year first above written.

Agreed to:

INTERNATIONAL BUSINESS
MACHINES CORPORATION, as Assignor

By [Signature]
Name ROBERT BELDEN
Title VP BUSINESS DEVELOPMENT

Date 10-28-15

Agreed to:

GLOBALFOUNDRIES U.S. 2 LLC, as Assignee

By [Signature]
Name Louis M. Lupin
Title Chief Legal Officer

Date 10-28-15

Connecticut
State of ~~New York~~)
) S.S. Brookfield
County of ~~Westchester~~
 Fairfield

Before me this 28 day of October, 2015 personally appeared Robert Belden Jr.
to me known to be the person who is described in and who signed the foregoing Assignment and acknowledged
to me that he/she signed the same of his/her own free will for the purpose therein expressed.

[Signature]

My Commission Expires
Notary Public October 31, 2020

State of New York)
) S.S.
County of [unclear]

Before me this 1st day of January, 2016, personally appeared Louis M. Lupin
to me known to be the person who is described in and who signed the foregoing Assignment and acknowledged
to me that he/she signed the same of his/her own free will for the purpose therein expressed.

[Signature]

Notary Public

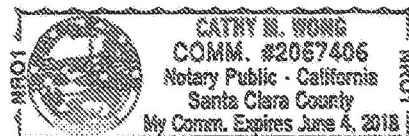


EXHIBIT A

Patents (United States)

Country	Docket Number	Patent Number	Patent Issue Date
US	FIS920130119US1	9082851	07/14/15
US	YOR920140089US1	9041079	05/26/16
US	FR919870016US1	5166552	11/24/92
US	FIS920130042US2	9059139	06/16/15

END OF EXHIBIT A

EXHIBIT B

Patents (Foreign/Non-US)

Country	Docket Number	Patent Number	Patent Issue Date
DE	FIS920060122DE1	2047506	08/08/15
DE	FIS920060030DE1	2024995	07/01/15
GB	FIS920060122GB1	2047506	08/08/15
GB	FIS920060030GB1	2024995	07/01/15
GB	ROC920130085GB1	2520597	05/27/15
CA	FR919870016CA1	1313401	02/02/93
CH	FR919870016CH1	318624	11/10/93
DE	FR919870016DE1	3788132908	11/10/93
ES	FR919870016ES1	318624	11/10/93
FR	FR919870016FR1	318624	11/10/93
GB	FR919870016GB1	318624	11/10/93
JP	FR919870016JP1	2538660	07/08/96
NL	FR919870016NL1	318624	11/10/93

END OF EXHIBIT B

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EXHIBIT C

Patent Applications (United States and Foreign/Non-US)

Country	Docket Number	Application Serial Number	Filing Date
CN	YOR920120827CN1	201380063493.7	08/16/13
CN	YOR920120088CN1	201380065013.0	08/28/13
CN	YOR920120171CN1	201380064097.6	12/06/13
DE	YOR920120827DE1	112013005369.1	12/05/12
DE	YOR920120088DE1	112013006010.8	08/28/13
DE	YOR920120171DE1	112013005871.5	12/06/13
GB	FIS920120247GB1	1509409.7	08/15/13
GB	YOR920120827GB1	1510917.6	08/16/13
GB	YOR920120088GB1	1511021.6	08/28/13
GB	YOR920120171GB1	1511020.8	12/06/13
KR	FIS920120269KR1	2015-7015655	02/17/14
US	YOR920130051US3	14/658898	03/16/15
US	YOR920130365US2	14/710204	05/12/15
US	SJO920130150US4	14/711119	05/13/15
US	YOR920130401US3	14/711482	05/13/15
US	YOR920130595US3	14/717344	05/20/15
US	YOR920110383US2	14/717387	05/20/15
US	YOR920100063US2	14/717551	05/20/15
US	YOR920130425US3	14/720427	05/22/15
US	YOR920120581US4	14/720328	05/22/15
US	YOR920120581US5	14/720393	05/22/15
US	YOR920130446US3	14/721840	05/26/15
US	FIS920120093US3	14/722411	05/27/15
US	YOR920110190US3	14/723095	05/27/15
US	FIS920120159US2	14/723681	05/28/15
US	BUR920120031US2	14/723703	05/28/15
US	YOR920130426US3	14/725151	05/29/15
US	YOR920120406US4	14/725795	05/29/15
US	YOR920120406US3	14/725755	05/29/15
US	YOR920120628US3	14/725581	05/29/15
US	AUS920150045US2	14/729177	06/03/15
US	YOR920040056US4	14/729446	06/03/15
US	AUS920130178US2	14/730375	06/04/15
US	YOR919980510US5	14/730503	06/04/15
US	FIS920130287US2	14/731569	06/05/15
US	YOR920120058US2	14/732680	06/06/15
US	YOR920130180US3	14/732689	06/06/15
US	YOR920130428US2	14/732705	06/06/15
US	BUR920130046US2	14/732835	06/08/15
US	BUR920110124US2	14/733235	06/08/15
US	YOR920110023US2	14/733652	06/08/15
US	BUR920120151US2	14/734713	06/09/15
US	BUR920140003US2	14/734310	06/09/15
US	YOR920130232US2	14/734018	06/09/15
US	FIS920130024US2	14/735466	06/10/15
US	FIS920110183US5	14/736698	06/11/15
US	FIS920090054US5	14/736651	06/11/15

US	YOR920130800US3	14/736942	06/11/15
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US	SJO920130045US3	14/738025	06/12/15
US	YOR920090221US5	14/739669	06/15/15
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US	FIS920080285US4	14/739703	06/15/15
US	YOR920130385US2	14/739137	06/15/15
US	YOR920090221US3	14/739562	06/15/15
US	END920130118US2	14/739262	06/15/15
US	YOR920090221US6	14/739886	06/15/15
US	YOR920120076US2	14/741169	06/16/15
US	FIS920110204US2	14/741618	06/17/15
US	FIS920120143US2	14/746017	06/22/15
US	BUR920130172US2	14/747525	06/23/15
US	YOR920121032US4	14/749772	06/25/15
US	YOR920121032US3	14/749731	06/25/15
US	TUC920130063US2	14/750971	06/25/15
US	YOR920130740US8	14/751761	06/26/15
US	YOR920130740US4	14/751493	06/26/15
US	YOR920130740US6	14/751646	06/26/15
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US	YOR920130740US7	14/751706	06/26/15
US	AUS920120164US2	14/754585	06/29/15
US	CAM920090008US2	14/754013	06/29/15
US	YOR920110024US3	14/754190	06/29/15
US	BUR920110204US2	14/755522	06/30/15
US	BUR920130114US2	14/755862	06/30/15
US	BUR920130130US2	14/788945	07/01/15
CN	ROC920130197CN1	201410645818.9	11/12/14
CN	DE920130113CN1	201410641489.0	11/13/14
CN	FIS920130119CN1	201410671056.X	11/21/14
CN	ROC920130060CN1	201410679750.6	11/24/14
CN	CHA920130028CN1	201410680044.3	11/24/14
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CN	IN920130077CN1	201410685922.0	11/25/14
CN	DE920130080CN1	201410687013.0	11/25/14
DE	END920130133DE1	102014116367.2	11/10/14
GB	GB920130113GB1	1320538.0	11/21/13
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GB	CH920130060GB1	1320982.0	11/28/13
PCT	FIS920130205PCT1	CN2014/088537	10/14/14
PCT	FIS920130121PCT1	CN2014/089293	10/23/14
TW	TW920130034TW1	102143309	11/27/13
TW	FIS920130205TW1	103121862	06/25/14
US	SJO920130134US1	14/081675	11/15/13
US	CA920130097US1	14/081946	11/15/13
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US	BUR920150025US1	14/733445	06/08/15
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US	FIS920140281US1	14/738288	06/12/15
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US	FIS920150099US1	14/749843	06/25/15
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US	BUR920150059US1	14/753771	06/29/15
US	POU920150093US1	14/753682	06/29/15